

OFFICIAL SCRUM TESTING VERIFICATION

Release v1.0.0

NatureSync — System Test Automation Report

Automated End-to-End, Machine Learning Model, and Web Portal UI Verification via Selenium WebDriver

TOTAL TESTS

7

Automated Test Cases

PASSED

7

100% Meets Criteria

FAILED

0

Zero Regressions

SUCCESS

RATE

100.0%

Verified & Validated

TOTAL

DURATION

58.09s

Executed on 13 Sep 2026, 01:49:24 PM

Operating System: Windows 11 (AMD64)

Python Runtime: 3.12.0

Automation Tool: Selenium WebDriver (Chrome)

Scope: Local Offline / Isolated Environment

Detailed Test Execution Matrix

Comprehensive trace for every module test case and verification criteria

TEST ID	MODULE & TEST NAME	OBJECTIVE	INPUTS	EXPECTED OUTPUT
TC-01	Backend Health Check & Service Gateway API Gateway	Verify that the local FastAPI server is online, healthy, and responsive	GET http://localhost:8000/health	HTTP 200 OK with {"status": "ok"}
TC-02	Interactive FastAPI Swagger UI Automated Exploration API Documentation	Verify Swagger UI loads correctly with all router tags (Crop, Disease, Market, Agent, Farm)	URL: http://localhost:8000/docs	Swagger UI displays title 'Smart Farming Assistant API' with interactive endpoint tags

TEST ID	MODULE & TEST NAME	OBJECTIVE	INPUTS	EXPECTED OUTPUT
TC-03	ML Crop Recommendation Algorithm Inference Machine Learning (Crop AI)	Verify that the ML model produces valid crop recommendations and confidence metrics from soil inputs	N=90, P=42, K=43, Temp=20.8°C, Hum=82%, pH=6.5, Rain=202.9mm	Valid recommended crop (e.g. 'rice') with confidence score > 0.70
TC-04	LeafDoctor Vision AI Disease Detection & Treatment Protocol Vision AI (LeafDoctor)	Verify diagnostic pipeline for Kerala crop diseases (Pepper Wilt, Banana Bunchy Top, Healthy)	Target: Black Pepper Quick Wilt (Phytophthora capsici)	Diagnostic identification with organic treatment guidelines (1% Bordeaux mixture)
TC-05	Kerala Mandi Market Price Intelligence Tracker Market Intelligence	Verify commodity market rates, arrival volumes, and modal prices for Kerala cash crops	District: Wayanad / Kerala; Commodity: Black Pepper & Rubber RSS-4	Valid commodity list with modal price (Rs/Quintal or Rs/Kg) and market arrival trends
TC-06	Agentic AI Conversational Farm Advisor Agentic AI (Advisory)	Verify agricultural reasoning engine for crop planning, monsoon risk, and pest control	Prompt: 'What should I intercrop with black pepper in Wayanad during monsoon?'	Grounded agronomic guidance recommending ginger/coffee with soil drainage advice
TC-07	End-to-End Farmer Web Portal UI Automation Web UI & User Experience	Verify that all 6 core portal views, language toggle, diagnosis modals, charts, and AI chat operate smoothly in Chrome	File: naturesync_farm_assistant.html	All navigation transitions, language changes, modal overlays, and charts load without JavaScript errors

Automated Visual Evidence Gallery

High-resolution browser snapshots captured automatically by Selenium during live execution

Smart Farming Assistant API 0.1.0 OAS 3.1
Recommendation, disease detection, weather, market intelligence, and AI-powered agricultural advisory.

Recommendation

1/crop-recommendation Get Crop Recommendation

1/agent/sessions Open Chat Session

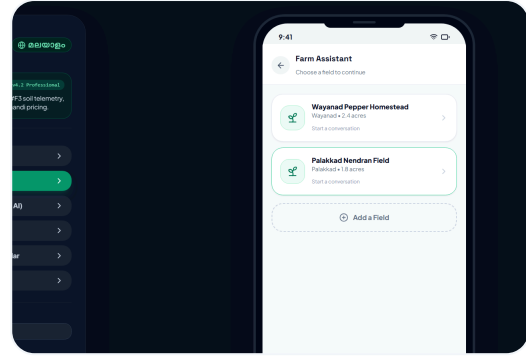
1/agent/crop-advisory Crop Advisory

*Farms Get My Farms

TC-02

VERIFIED

Interactive FastAPI Swagger UI Automated...



TC-07

VERIFIED

End-to-End Farmer Web Portal UI Automation

Scrum Milestone Sign-off & Faculty Review

Official sign-off section for project diary and academic sprint evaluation

STUDENT / PROJECT TEAM DECLARATION

I hereby declare that the automated test cases above have been executed successfully on the NatureSync codebase using Selenium WebDriver. All core modules (ML Crop Engine, Vision AI Doctor, Mandi Prices, and UI) satisfy their defined acceptance criteria.

Student Developer Signature

Date: 13 Sep 2026

FACULTY COORDINATOR / GUIDE APPROVAL

This test execution report has been evaluated. The automated test runs, pass percentage (100.0%), and photographic proofs satisfy the Scrum book verification criteria for this milestone.

Faculty Coordinator Signature

Status: Approved

NatureSync AI-Powered Smart Farming Assistant • Generated with Selenium WebDriver Test Suite